

REVIEW ARTICLE



Microbiome, GvHD, and immune reconstitution in allogeneic hematopoietic cell transplantation

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The gut microbiota has emerged as a critical factor influencing outcomes following allogeneic hematopoietic cell transplantation (alloHCT). Notably, disruptions to the intestinal microbiome—referred to as dysbiosis—have been strongly linked to the development of acute graft-versus-host disease (aGVHD). The gut microbiome interacts closely with the host immune system, influencing both immune reconstitution and alloHCT complications. As a result, microbiome-targeted strategies are being investigated to improve outcomes and include antibiotic stewardship, prebiotic and diet intervention, probiotics including fecal microbiota transfer (FMT) and postbiotics. These approaches are being investigated not only as a therapeutic intervention in particular for aGVHD, but also as preventive strategies.

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INTRODUCTION

Human microbiota, defined as the community of microorganisms that colonize humans and their collective genomes (the microbiome), is in close relationship with health and disease. Importantly, microbiota and host coevolve from birth, leading to a unique host–microbiome symbiosis in healthy individuals [1]. The 100 trillion bacteria of the human microbiota interact daily with the skin, mouth, pharynx, respiratory system, urogenital tract, and the digestive tract. The latter contains by far the largest density and diversity of microorganisms. Gut microbiota dysbiosis, the imbalance in the normal composition, diversity, or function of the gut microbial communities, is associated with the disruption of the host–microbiome symbiosis, that leads to alteration of the gut barrier (most often permeability) [2] and disruption of immunologic homeostasis. In the setting of hematological malignancies this is particularly relevant since those patients often experience numerous insults to the microbiome, such as impaired nutritional intake and exposure to chemotherapy and broad-spectrum antibiotics, that can lead to dysbiosis, disruption of symbiosis and local and systemic inflammation [3]. In the last decade, numerous studies investigated the interplay between gut microbiome dysbiosis and outcomes of patients with hematological malignancies. It became clear that there is a close relationship between the gut microbiota and outcome after allogeneic hematopoietic cell transplantation (alloHCT) [4–6].

Acute GvHD is an immunological complication of alloHCT, associated with high morbidity and mortality, particularly when the gastrointestinal (GI) tract is affected [7]. In the early 1970s and 1980s, data in murine models and later in clinical studies suggested that intestinal microbiota play an important role in the pathophysiology of acute GvHD [8, 9]. It was demonstrated

that alloHCT in germ-free mice as well as total or selective decontamination with broad-spectrum antibiotics resulted in a significant reduction of GvHD-related mortality [8, 9]. Subsequent experimental studies showed that manipulation of the intestinal microbiota composition either by antibiotics such as Ampicillin or by introducing potentially beneficial bacteria such as *Lactobacillus* significantly modified intestinal inflammation and outcome of transplant recipients [5, 10]. The progress of the research was however impeded by the absence of methodologies to culture the whole spectrum of anaerobic and aerobic bacteria. The recent introduction of culture-independent techniques involving ribosomal RNA (rRNA) gene sequencing finally enabled us to untangle the complex relationship between gut microbiota, bacterial-derived metabolites and host tissues and their significant impact on immune cells, immune reconstitution and GvHD pathogenesis [11] the topic of this review.

RISK FACTORS IMPACTING MICROBIOME DYSBIOSIS AND IMPACT OF MICROBIOME ON PATIENTS' OUTCOME AFTER ALLOHCT

Before and after alloHCT, the diversity of the intestinal microbiome significantly decreases over time. Disruption of the homeostatic microbiota composition and, particularly, early intestinal dysbiosis, is strongly associated with poor outcome [6, 12, 13]. In a large-scaled multicenter study, similar microbial patterns characterized by loss of commensals and domination by single bacterial taxa such as *Enterococci* were associated with high transplant-related mortality, independent of the institutes' geographic locations and practices [6, 14]. Intestinal dysbiosis creates a pro-inflammatory milieu and leads to disruption of epithelial integrity. Epithelial

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damage is exacerbated by pretransplant conditioning, which causes the release of intracellular molecules, damage-associated molecular patterns (DAMPs), while activation or destruction of bacteria induces the release of pathogen-associated molecular patterns (PAMPs) [7]. DAMPs and PAMPs act as danger signals to activate antigen presenting cells, which are then responsible for the activation of allogenic donor lymphocytes and acute GVHD [7]. Epithelial damage, bacterial translocation and inflammation are the hallmarks of aGVHD development and at the origin of blood stream infections and septicemias.

Alteration of the intestinal microbiota is caused by antibiotics given either during leukemia remission-induction therapy or after alloHCT, by dietary changes, by pre-transplantation conditioning therapy and by the use of other medications. The majority of alloHCT recipients is extensively exposed to prolonged prophylactic and therapeutic antibiotics during episodes of neutropenia. These are often broad-spectrum antibiotics that target obligate anaerobic bacteria, typically producers of short-chain fatty acids (SCFA) in the gut. Depletion of SCFA producing bacteria is associated with expansion of opportunistic pathogens [12, 13, 15]. In addition, absence of normal dietary intake and prolonged parenteral nutrition leads to loss of protective beneficial bacteria such as *Blautia* [16–18]. Finally, conditioning regimens including chemotherapy and radiation, and immuno-suppressive medications to prevent GVHD may disrupt the sensitive equilibrium between host and microbiota [19–22].

INTERPLAY BETWEEN MICROBIOME, IMMUNE CELL SUBSETS AND GUT-DERIVED METABOLITES AFTER ALLOHCT

Intestinal microbiota are able to modify local and systemic immunity either by direct interaction with immune cells or indirectly via the production of microbial metabolites such as SCFA like butyrate, bile acids, polyamines, and indoles binding to the aryl hydrocarbon receptor (AHR) [23].

Crosstalk between microbiota and immune cell subsets

The gut microbiome actively shapes immune reconstitution following alloHCT by regulating the balance between regulatory T cells (Tregs), pro-inflammatory T helper 17 (Th17) cells—the predominant CD4⁺ T cell subsets in the lamina propria under steady-state conditions and innate lymphoid cells (ILC). Microbial diversity, specific bacterial taxa, and metabolite production play crucial roles in maintaining this balance, influencing both GVHD outcomes and immune recovery.

Tregs play a pivotal role in maintaining immune homeostasis and promoting recovery, particularly in the context of gut microbiome interactions. Specific commensal bacteria, such as *Bacteroides fragilis*, further amplify Treg development via polysaccharide A (PSA)-dependent signaling, which primes dendritic cells to foster Foxp3⁺ Treg induction [24]. Recent findings highlight that these microbiota-induced Tregs not only expand in number but also critically interact with alloreactive CD8⁺ T cells to directly suppress their allogeneic activation, thereby limiting the severity of GVHD. Disruption of the microbiome leads to a marked reduction in Tregs and their regulatory crosstalk with alloreactive cytotoxic T cells, resulting in heightened GVHD severity [25].

Th17 cells are a subset of pro-inflammatory T helper cells characterized by their production of interleukin 17 (IL-17). Th17 play a key role in host resistance against intestinal pathogens and promote systemic autoimmunity [26, 27]. Certain commensal bacteria, including segmented filamentous bacteria (SFB) and *Bifidobacterium adolescentis*, are known to drive Th17 differentiation [28]. Studies in both human and murine models indicate that higher levels of IL-17-producing donor-derived CD4⁺ T cells are detected in cases of severe intestinal GVHD, where IL-17 enhances tissue injury by promoting neutrophil infiltration and macrophage

activation [25, 29]. Microbiota-derived signals such as SFB-induced IL-6 and IL-23 pathways further skew the Th17/Treg balance toward inflammation, while loss of microbial diversity reduces regulatory feedback mechanisms that normally limit Th17 expansion. Certain microbial metabolites, such as 3-oxo LCA (a secondary bile acid), can inhibit Th17 cell differentiation and mitigate GVHD, suggesting that both the composition and metabolic function of the microbiota directly affect the Treg/Th17 axis and GVHD outcomes [30].

ILCs are recently discovered heterogenous group of innate immune cell specialized in the rapid secretion of cytokines and chemokines to combat infection and promote mucosal tissue repair [31]. ILC3s, a subset of ILCs, play a key role in gut immunity, and their function is regulated by Ffar2, a receptor for microbial metabolites [32]. Ffar2 activation enhances IL-22 production, which strengthens the gut barrier and promotes epithelial healing. Additionally, ILC3s suppress alloreactive T cell activation [33], help monitor gut microbiota, and support early colonization resistance through ID2-dependent IL-22 regulation [34]. In murine alloHCT models, ILC3-derived IL-22 protects the intestinal stem cells, limits epithelial damage, and enhances production of antimicrobial peptides and mucins, all promoting effective epithelial repair and reduced severity of GVHD [19].

Other important immune cells also contribute to immune reconstitution and their interplay with the microbiota and its derived metabolites are summarized in Table 1.

Gut-derived metabolites and immune reconstitution

Short-chain fatty acids (SCFA). The gut microbiota enhances immune tolerance by fermenting dietary polysaccharide SCFA like butyrate, propionate, and acetate, which have key immunomodulatory effects. Butyrate regulates hematopoiesis by activating toll-like receptor signaling in hematopoietic stem cells and contributes to the overall restoration of immune cell numbers [35]. Butyrate produced by *Clostridium* clusters (IV, XIVa, XVIII) and *Bifidobacterium*, enhance Treg differentiation by inhibiting histone deacetylases (HDACs) and maintaining Foxp3 expression, thereby expanding colonic Treg populations, which are essential for reducing inflammation and promoting mucosal repair [36]. Consistent with these findings, vorinostat, a pharmacological HDAC inhibitor, has been shown to decrease inflammatory cytokines, lower established GVHD biomarkers (such as Reg3 and ST2), and enhance Treg function, thereby reducing the incidence and severity of acute GVHD in both preclinical and clinical studies [37, 38].

Bile acids. Bile acids, synthesized by the liver and modified by gut microbiota into secondary forms (e.g., lithocholic acid (LCA), deoxycholic acid (DCA)), regulate immune responses via receptors like farnesoid X receptor (FXR) and G-Protein Bile Acid-activated Receptor (GPBAR). Secondary bile acids, such as 3-oxoLCA and isoalloLCA, modulate the Th17/Treg balance: 3-oxoLCA inhibits ROR γ t-driven Th17 differentiation, while isoalloLCA enhances FOXP3⁺ Treg induction [30]. Prophylactic tauroursodeoxycholic acid (TUDCA) administration reduced acute GVHD severity in murine transplantation models by decreasing antigen presentation activity and preventing intestinal epithelial apoptosis, independent of microbiome alterations [39].

Tryptophan metabolites. Tryptophan, an essential amino acid derived primarily from dietary sources, is metabolized via three interconnected pathways: kynurenine, serotonin, and indole pathways. In the context of allo-HSCT, microbiota-derived tryptophan metabolites play a crucial role in immune recovery. Indole derivatives produced by commensals like *Lactobacilli* and *Proteobacteria* activate the aryl hydrocarbon receptor (AHR), which regulates the IL-22/IL-10 axis, promotes Treg differentiation and ILC homeostasis, and suppresses hyperinflammation [40].

Table 1. Interactions between immune cells and gut microbiota.

Immune Cell Type	Crosstalk with Gut Microbiota	Key Metabolites/ Mechanisms	Functional Impact	Therapeutic Relevance	References
Macrophages	<i>Bacteroides fragilis</i> enhances phagocytic function via polysaccharide A (PSA). SCFAs (e.g., butyrate) reduce NF- κ B signaling and promote anti-inflammatory M2 macrophages. LPS from <i>Enterobacteriaceae</i> induces pro-inflammatory M1 polarization.	SCFAs (butyrate, propionate), PSA, LPS	Balances inflammation; promotes tissue repair and pathogen clearance.	FMT to restore SCFA-producing taxa; engineered probiotics targeting M2 polarization.	[30, 90]
Dendritic Cells (DCs)	SFB (<i>Segmented Filamentous Bacteria</i>) induce Th17 differentiation via IL-23. <i>Lactobacillus</i> spp. prime tolerogenic DCs through AhR ligands (e.g., indole derivatives).	AhR ligands (indole derivatives), flagellin	Enhances antigen presentation; balances Th17/Treg responses; maintains intestinal homeostasis.	AhR agonists (e.g., tryptophan metabolites); microbial consortia targeting DC maturation.	[28, 90]
NK Cells	<i>Akkermansia muciniphila</i> enhances NK cytotoxicity via TLR2 signaling. Dysbiosis reduces NK cell-mediated tumor surveillance.	TLR2 ligands, SCFAs	Improves anti-tumor immunity; maintains mucosal barrier integrity.	<i>Akkermansia</i> supplementation; TLR2 agonists.	[91]
B Cells	Microbiota-driven IgA class-switching via Peyer's patches. <i>Bifidobacterium</i> spp. promote IL-10+ regulatory B cells (Bregs).	IgA, polysaccharide A (PSA), SCFAs	Neutralizes pathogens; reduces systemic autoimmunity; promotes gut homeostasis.	IgA-boosting probiotics; engineered IgA-coated microbes.	[92]
CD8+ T Cells	Butyrate inhibits HDACs, enhancing memory CD8+ T cell differentiation. Dysbiosis exacerbates CD8+ exhaustion via PD-1 upregulation.	Butyrate, isoDCA, PD-1 ligands	Enhances anti-tumor cytotoxicity; mitigates exhaustion in chronic infections.	Butyrate supplements; PD-1/PD-L1 inhibitors.	[92, 93]
Th17 Cells	SFB and <i>Citrobacter rodentium</i> drive ROR γ t+ Th17 cells via IL-6/IL-23. SCFAs and lithocholic acid derivatives (e.g., 3-oxoLCA) suppress pathogenic Th17 differentiation by inhibiting ROR γ t activity.	SCFAs, 3-oxoLCA	Protects against fungal infections; exacerbates GVHD if dysregulated.	ROR γ t antagonists; SFB-targeted antibiotics.	[26, 27]
Treg Cells	<i>Clostridia</i> clusters produce butyrate, expanding colonic Foxp3+ Tregs. <i>Bacteroides fragilis</i> PSA induces IL-10+ Tregs via TLR2. Secondary bile acids like isoalloLCA promote mitochondrial ROS for Foxp3 expression. Dysbiosis and antibiotics reduce Treg numbers and alter their crosstalk with CD8 T cells, tipping balance toward GVHD.	Butyrate, PSA, isoalloLCA, SCFAs (butyrate), tryptophan catabolites, secondary bile acids.	Suppresses GVHD; maintains intestinal tolerance; enhances mucosal repair; suppresses alloreactive CD8 T cell proliferation and tissue damage.	Postbiotics (e.g., butyrate analogs); FMT for Treg expansion; microbiome-preserving therapies may maintain Treg-CD8 balance	[24, 25, 36]

Table 1. continued

Immune Cell Type	Crosstalk with Gut Microbiota	Key Metabolites/Mechanisms	Functional Impact	Therapeutic Relevance	References
ILC3s	<i>Alistipes</i> spp. activate AhR in ILC3s to produce IL-22. Dysbiosis reduces ILC3-mediated epithelial repair and antimicrobial peptide production.	AhR ligands (indole derivatives), SCFAs	Enhances mucosal barrier repair; regulates lymphoid tissue development.	AhR agonists (e.g., dietary indoles); <i>Alistipes</i> consortia.	[31, 34]
$\gamma\delta$ T Cells	Microbiota-derived D-lactate primes $\gamma\delta$ T cells for IL-17 production. <i>Lactobacillus</i> spp. modulate $\gamma\delta$ T cell trafficking to gut mucosa through GPR15 signaling.	D-lactate, SCFAs	Early defense against pathogens; promotes tissue homeostasis.	D-lactate-producing probiotics; $\gamma\delta$ T cell boosters.	[94]

AhR aryl hydrocarbon receptor, DCs dendritic cells, *FMT* fecal microbiota transplantation, *Foxp3* forkhead box P3, *GPR15* G protein-coupled receptor 15, *GVHD* graft-versus-host disease, *HDAC* histone deacetylase, ILC3s type 3 innate lymphoid cells, *IL* interleukin, *isoDCA* iso-deoxycholic acid, *isoalloLCA* isoallo-lithocholic acid, *LPS* lipopolysaccharide, *M1/M2* classically/alternatively activated macrophage phenotypes, *NF- κ B* nuclear factor kappa-light-chain-enhancer of activated B cells, *NK* cells, natural killer cells, *PD-1* programmed cell death protein 1, *PD-L1* programmed death-ligand 1, *PSA* polysaccharide A, *ROryt* retinoic acid-related orphan receptor gamma t, *SCFAs* short-chain fatty acids, *SFB* segmented filamentous bacteria, *TLR* toll-like receptor, *Tregs* regulatory T cells.

Conversely, the kynurenine pathway, mediated by indoleamine 2,3-dioxygenase (IDO), contributes to immunosuppression by depleting tryptophan and accumulating kynurenines that promote Treg expansion while impairing effector T cells. Dysbiosis post-transplant alters tryptophan metabolism, reducing AhR ligands and exacerbating GVHD.

Myeloid-derived suppressor cells (MDSCs) are also key mediators of this pathway, playing an essential role in maintaining immune tolerance after allo-HSCT. MDSCs express IDO, which catalyzes tryptophan degradation into immunosuppressive kynurenines that activate AhR signaling. This reciprocal interaction promotes MDSC expansion and enhances their ability to suppress alloreactive T cell responses, thereby attenuating GvHD severity while preserving graft-versus-leukemia effects. In addition, MDSCs release arginase-1 and nitric oxide, further inhibiting T cell proliferation and supporting Treg induction. Together, these mechanisms define the tryptophan–MDSC axis as a central pathway balancing immune tolerance and inflammation in GvHD [41].

MICROBIOME AND IMMUNE RECONSTITUTION AFTER ALLOHCT

Immune reconstitution following alloHCT sets the stage for a successful transplant. A properly reconstituted immune system requires both innate and adaptive arms for defense against viral and microbial pathogens, tissue repair, and beneficial graft-vs-leukemia effects. Immune reconstitution is dependent on factors including conditioning regimen, graft source and manipulation, transplant outcomes such as GVHD, and fixed patient factors like age-related thymic function. For adaptive immunity, numerical T and B cell recovery typically occurs between 6 to 12 months, while full reconstitution may take up to 2 years [6, 42]. In parallel, innate immunity reconstitutes along distinct timelines: NK cells and subsets of ILCs (ILC1 and ILC3) recover within weeks, whereas ILC2 may take years [43]. ILCs play key role in epithelial tissue repair and maintain a reciprocal relationship with the microbiome which is essential for their homeostasis in the gut [44].

Emerging evidence is now revealing that the gut microbiota can also impact immune recovery post-alloHCT in murine models and in patients (Table 2). Higher fecal microbial diversity in the early post-transplant periods strongly associated with improved CD4⁺ T-cell recovery, particularly in patients receiving CD34-selected grafts and who rely on de novo lymphopoiesis [45]. In contrast, low diversity correlates with increased GVHD, infections, and mortality.

In addition, specific colonization patterns and microbial nutrient interactions have been associated with immune deregulation. Mono-domination, where a single taxon comprises more than 30% of the microbial community, is a common feature during such periods, particularly during phases like perineutrophil engraftment [6]. Dietary lactose has been identified as a key driver of Enterococcus expansion, with lactose malabsorption further impairing microbial clearance [46]. Clinically, early dominance of Enterococcus correlates with CD4⁺ lymphopenia and increased risk of bloodstream infections. In contrast, high abundance of beneficial commensals such as *Blautia* has been associated with accelerated recovery of CD4⁺ and CD8⁺ T cells, enhanced antiviral immunity, and reduced risk of leukemia relapse [47]. Moreover, the presence of butyrate-producing taxa—including members of Lachnospiraceae and Ruminococcaceae—has been associated with robust B cell reconstitution and intestinal IgA production, key to mucosal defense [42].

MICROBIOME AND GVHD

The impact of bacterial metabolites that are normally involved in epithelial repair and immunological regulation on the

Table 2. Clinical and preclinical studies on microbiota-immune reconstitution interactions.

Study	Cohort/Model	Microbial Focus	Immune Metrics	Mechanism/Findings	Clinical Impact	Limitations
Taur et al. [12]	Humans (allo-HSCT, n = 80)	Alpha diversity via Inverse Simpson index	Stem cell source, infection, and microbial diversity	Increased OS associated with higher relative abundance of <i>Lachnospiraceae</i> and <i>Actinomycetaceae</i> . Domination by <i>Enterococcus</i> , <i>Streptococcus</i> , <i>Enterobacteriaceae</i> , and <i>Lactobacillus</i> spp. linked to poor outcomes.	Replenishment of commensal bacterial populations may improve immune reconstitution and outcomes post-HSCT.	Causal relationships unclear; control interventional studies needed.
Iwamura et al. [95]	Murine model (NOD1 – / – , SPF, GF)	Microbiota in the maintenance of steady-state hematopoiesis	NOD1 and NOD2 ligands	NOD1 ligand administration restores hematopoietic precursor pools in germ-free mice to levels seen in specific pathogen-free animals.	Suggests NOD1 ligands as potential therapeutic agents to restore hematopoiesis post-HSCT.	Requires further studies to find the interaction between immune pathways
Staffas et al. [96]	Murine model (antibiotic-treated allo-HSCT)	Antibiotics, sucrose metabolism	Granulocyte/lymphocyte recovery	Depletion of intestinal microbiota impairs dietary energy uptake and visceral fat store. Caloric supplementation via sucrose restored granulopoiesis post-allo-HSCT.	Suggests postbiotics (e.g., sucrose) as potential therapies for neutrophil recovery post-HSCT.	Preclinical model; relevance to human immune reconstitution requires further validation.
Peled et al. [6]	Humans (allo-HSCT, n = 1,362)	Loss of microbial diversity and domination of single taxa	OS, neutrophil engraftment, GVHD	Higher diversity of intestinal microbiota at the time of neutrophil engraftment was associated with increased OS. Abundance of <i>Enterococcus</i> and <i>Streptococcus</i> spp. linked to increased GVHD risk.	Identifies microbial diversity as a biomarker for post-HSCT outcomes.	Retrospective design; heterogeneity in conditioning regimens.
Schluter et al. [42]	Humans (allo-HSCT, n = 841; auto-FMT, n = 24)	Microbial community dynamics	Daily counts of neutrophils, lymphocytes, and monocytes	Higher counts of WBCs in individuals receiving auto-FMT in the first 100 days after neutrophil engraftment. Enriched microbiota with <i>Faecalibacterium</i> , <i>Ruminococcus</i> 2, <i>Akkermansia</i> accelerates immune reconstitution.	Identifies specific taxa that predict immune kinetics during the engraftment phase post-HSCT.	Lacks detailed information on lymphocyte subtypes and other immune cell populations.
Miltiadows et al. [97]	Humans (allo-HSCT, n = 894)	Early post-HSCT microbiome (d7–21)	Day 100 CD4+ / CD8+ T cells, B cells, NK cells	Increased abundance of <i>Staphylococcus</i> associated with poor CD4 T-cell recovery. Highlights <i>Erysipelotrichaceae</i> as beneficial for immune reconstitution.	Identifies key microbiome taxa associated with immune recovery post-HSCT.	Observational study; heterogeneity in GVHD prophylaxis regimens.
Thiele Orberg et al. [53]	Humans (allo-HSCT, n = 78)	Bacteria and bacteriophage consortia	SCFA biosynthesis pathways, survival outcomes	High SCFA-producing pathways in bacteria/phages associated with improved survival and reduced relapse post-HSCT.	Supports the development of synthetic metabolite formulations as adjunctive therapies post-allo-HSCT.	Requires validation in larger cohorts.

allo-HSCT allogeneic hematopoietic stem cell transplantation, *CD* cluster of differentiation, *FMT* fecal microbiota transplantation, *GF* germ-free, *GVHD* graft-versus-host disease, *NOD* nucleotide-binding oligomerization domain, *OS* overall survival, *SCFA* short-chain fatty acid, *SPF* specific pathogen-free, *WBC* white blood cell.

pathogenesis of GvHD proceeds through direct and indirect mechanisms. They can induce anti-inflammatory responses [29], improve intestinal epithelial barrier integrity, help maintain intestinal homeostasis and they can enhance anti-inflammatory Treg and tissue-repair inducing ILC3 [25, 48–51]. Intestinal commensals confer colonization resistance against enteropathogens by nutrient limitation and the production of bacteriocins and as such contribute to balanced gut homeostasis [52]. Recent data identified a microbiome signature of *Lachnospiraceae* and *Oscillospiraceae* and their associated bacteriophages, that correlated with the production of immunomodulatory metabolites (IMMs) such as short chain and branch chain fatty acids and interferon- γ inducing metabolites. The presence of these IMMs strongly correlated with alloHCT outcome [53].

Whereas the presence of SCFA-producing bacterial species was found to be associated with protection against acute GvHD-related mortality and the development of chronic GvHD [29, 54], other bacterial taxa have been associated with more severe disease. Loss of microbiome diversity is often accompanied with a shift to *Enterococcus* domination, which is associated with an increased risk of severe GvHD and GvHD-related mortality [4, 6]. In murine models enterococcal administration aggravated GvHD severity, demonstrating causality. Indeed, a diet free of lactose, a substrate mainly utilized by enterococci, prevented enterococcal overgrowth and reduced GvHD severity [14]. Similarly, the administration of butyrate or butyrate producing bacteria mitigated GvHD severity and improved transplant outcome in murine GvHD [21].

These data suggest that the pathophysiology of acute GvHD may depend on a complex interplay among various bacterial species, their metabolites, intestinal homeostasis, barrier function and immune tolerance. The most sensitive time period to interfere is probably when dysbiosis has not yet led to immune dysregulation, most likely in the immediate pre- and post-transplant period. This is suggested amongst others by the impact of timing of antibiotics.

MICROBIOME-TARGETED STRATEGIES TO IMPROVE OUTCOMES

Antibiotic stewardship

Broad-spectrum antibiotics are an integral part of alloHCT care, and frequently used to treat febrile infections, a major complication especially in the post-transplant neutropenic period. Additionally, prophylactic antibiotics have been routinely used aiming total or selective gut decontamination primarily to prevent infectious complications during aplasia but also to protect patients from GvHD since early studies suggested that gut decontamination might reduce GvHD severity, improve survival and also reduce neutropenic fever [8, 9].

However, the use of broad-spectrum antibiotics is a double-edged sword, as it remains a major risk factor driving severe intestinal dysbiosis, which is clearly associated with poor overall survival and an increased risk for acute GvHD [12, 55]. Antibiotics like carbapenems and piperacillin/tazobactam that suppress commensal bacteria have been associated with increased GvHD-related mortality, whereas antibiotics with commensal and anaerobic sparing effects like cefepime or aztreonam were not [56]. In mice, treatment with imipenem-cilastatin led to a degradation of the protective mucus layer and disruption of the intestinal barrier function. This has been attributed to specific bacteria such as *Akkermansia muciniphila*, a mucus degrading bacterium [56]. Interestingly, not only the type of antibiotics in general but also the timing seems to be critical, as patients who received antibiotics prior to the day of transplantation showed loss of protective *Clostridia* and increased transplant related mortality, compared to patients with later initiation of antibiotic treatment [55].

Given the negative impact of early antibiotic use on transplant outcome, clinical strategies for antibiotic stewardship have been an area of active investigation aiming a more sophisticated use of antibiotics in the setting of alloHCT. At a single transplant unit, the change of antibiotic guidelines to a more restrictive use of antibiotics particularly in cases with a high likelihood of cytokine release syndrome (CRS), e.g., after anti-thymocyte globulin therapy, led to reduced antibiotic administration in general, more microbiome diversity and a trend towards a reduced incidence of severe GI GvHD, without an increase in infectious complications [57]. Thus, the development of strategies allowing better differentiation between fever as a consequence of CRS and fever due to real infection would be a valuable clinical approach for optimal antibiotic stewardship.

A group of innovative new substances including Beta-lactam cleaving enzymes, charcoal-based adsorbents or engineered probiotics depict another promising option to maintain microbiome integrity and to mitigate dysbiosis associated risks. These antibiotic neutralizing agents selectively decrease antibiotic concentrations in the gut without affecting their therapeutic efficacy allowing the use of highly effective antibiotics for the treatment of severe infections [58–62].

The clinical significance of prophylactic antibiotics during the alloHCT period is still an unsolved issue. After initial beneficial reports regarding GvHD severity and OS in a clinical study of the Essen group using gut decontamination with ciprofloxacin and metronidazole [9], these effects could not be confirmed in follow-up studies. In the meantime, there is an increasing number of smaller clinical studies showing no protective effects of prophylactic antibiotics on acute GvHD resulting in a heterogeneous use of prophylactic gut decontamination across transplant centers, as randomized prospective studies on antibiotic prophylaxis are still missing [63–65].

In summary, these data describe paradoxical antibiotic effects in alloHCT: broad-spectrum antibiotics induces luminal and mucosal dysbiosis with negative effects on epithelial barrier integrity, containment of pathobionts, and transplant outcome [66]. These data seem to be contradictory to the old data reporting a reduced incidence of GvHD in germfree mice and also to recent findings indicating that microbiota suppression by antibiotics can positively interfere with bacterial translocation. Intensive conditioning damages the intestinal epithelium and facilitates bacterial translocation into the intestinal wall, which is amplified after loss of epithelial barrier integrity due to T cell mediated apoptosis in GvHD patients - a hallmark of GvHD initiation [67, 68]. Antibiotics can either enhance or suppress T-cell mediated immunity, depending on the context: While dysbiosis diminishes T-cell regulation and increases GvHD susceptibility, suppression of translocated bacteria dampens T-cell activation through reduced co-stimulation mitigating GvHD. Thus, in alloHCT a more sophisticated timing of antibiotics may supplement antibiotic stewardship in general, and strategies preserving luminal and mucosal eubiosis but preventing translocation may be needed [66] allowing a better balance between infection reduction and microbiota damage.

Probiotics, prebiotics, and postbiotics

Beside antibiotic stewardship, others strategies are under investigation to preserve or repair the microbiome in patients undergoing alloHCT including prebiotic and dietary changes, probiotics and fecal microbiota transfer, and postbiotic supplements.

Prebiotics are indigestible carbohydrates selectively utilized by host microorganisms. Riwe et al. conducted a phase 2 study to evaluate the use of resistant potato starch (RPS) in 10 adults' patients that received alloHCT [69]. RPS is a form of potato starch that is practically not digested by human enzymes in the small intestine [70]. It passes through to the large intestine largely intact,

where it undergoes fermentation by the oral and intestinal microbiota. Consequently, it functions as a prebiotic, promoting a healthy gut microbiota and potentially contributing to benefits like increased production of SCFA [70]. RPS was given daily from day -7 to day 100. This pilot study established that administration of RPS was feasible and tolerated by patients and translated in significantly higher fecal butyrate levels when participants were on RPS than when they were not ($p < 0.0001$). Nevertheless, longitudinal changes in plasma metabolites compared to baseline were independent of RPS ($P < 0.0001$). Another resistant starch (RS), Amylofiber SH, (corn starch with high amylose content) was evaluated in combination with GFO, a prebiotics mixture comprising glutamine, fiber, and oligosaccharide commercially available in Japan [71]. In a prospective trial that included 49 patients and 142 matched historic controls, RS and GFO were given from pretransplantation conditioning to day 28 after alloHCT. Prebiotics intake was associated with maintenance of gut microbial diversity and preservation of butyrate-producing bacterial. In particular fecal butyrate concentration after alloHCT was maintained or even increased in the prebiotics group. This translated in a clinical benefit with a significantly lower cumulative incidence of gastro-intestinal grade II-IV GvHD. The observation that RS and GFO also reduced the incidence of aGVHD of the skin suggests a systemic effect. Finally, Andersen et al. reported a pilot study that compared standard enteral nutrition ($n = 10$) with prebiotic fibre enteral nutrition ($n = 20$) [72]. Microbial diversity declined in both groups, however, in the prebiotic group a significant increase of the relative abundance of *Lactobacillus_C rhamnosus* ($p = 0.022$) and decreased expression of antibiotic resistance genes was observed. Overall, the use of prebiotics shows promising results, but larger randomized studies are warranted to confirm these findings. Regarding specific dietary interventions in alloHCT, additional reports and studies are also needed.

Probiotics are live microorganisms which, when administered in adequate amounts, confer a health benefit on the host. Probiotics can consist of single symbiotic bacterial strains, of a consortium of selected strains, may refer to engineered probiotics (genetic or metabolic modification of probiotic microorganisms) or to a full ecosystem (fecal microbiota transfer, FMT). In the setting of alloHCT, FMT is the most widely investigated approach. FMT involves the transfer of fecal material from healthy donors to restore gut microbial diversity and function. Initially established as an effective treatment for *Clostridioides difficile* infections resistant to antibiotics [73], FMT has expanded into experimental applications for immune modulation, including overcoming resistance to immune checkpoint inhibitors (ICI), treatment of gastro-intestinal (GI) aGVHD and improving outcomes after alloHCT. In ICI-resistant melanoma patients, phase I clinical trials demonstrated that FMT combined with anti-PD-1 therapy led to partial or complete tumor responses in a subset of patients, highlighting its potential to reverse immunotherapy resistance [74].

In the setting of GI-aGVHD, effective and safe use of FMT was first reported by Kakhana et al. in 2016 in 4 patients with steroid-resistant (SR, $n = 3$) and steroid-dependent (SD, $n = 1$) GI-aGVHD, with 3 patients that achieve a complete response [75]. Those results were further confirmed by several case reports and small series [76]. In a prospective single arm study, Van Lier and colleagues used FMT in 15 patients with steroid refractory (SR; $n = 6$) or steroid dependent (SD; $n = 9$) GI-aGVHD, with 10 patients achieving a complete response [$n = 3$ (50%) in SR and $n = 7$ (77%) in SD] [77]. A phase 2a study evaluated a pooled allogeneic FMT product, MaaT013, in patients with SR GI aGVHD ($n = 24$) and with an expanded access program another 52 patients were evaluated. The day 28 overall response rate was 38% and 58% respectively [78]. In all studies stools from responding patients demonstrated higher microbiota richness and increased levels of beneficial bacteria, in particular butyrate producers, along with increased

levels of short-chain fatty acid and bile acids. A prospective phase 3 study evaluating MaaT013 in SR and ruxolitinib refractory GI aGVHD is ongoing (NCT04769895). Finally, DeFilipp et al. evaluated FMT combine with combination with systemic corticosteroids in 10 patients with high-risk treatment naïve GI aGVHD, with a CR rate at day 28 of 70% [79]. Significant increase in tryptophan metabolites and short-chain fatty acids were also observed within the first 7 days after FMT [79].

Other studies investigated FMT (autologous or allogeneic) to restore microbiome composition after neutrophil engraftment [80–82]. Those studies report significant improvements in microbial diversity and total white blood cell (WBC) counts during the early post-engraftment period compared to controls [81]. A randomized, double-blind, placebo-controlled trial found that FMT improved post-antibiotic recovery of microbiota diversity, restored several depleted obligate anaerobic commensals, and reduced the abundance of expanded genera like *Enterococcus*, *Streptococcus*, *Veillonella*, and *Dialister* [83]. Finally, Peled et al. reported as a congress abstract the preliminary results of a phase 1b study, evaluating SER-155, an oral investigational cultivated microbiome therapeutic comprised of 16 bacterial Firmicutes strains, given before conditioning and after neutrophil engraftment [84]. Compared to placebo ($n = 14$), SER-155 ($n = 20$) was generally well tolerated with a safety profile similar to placebo and the incidence of bloodstream infections, a secondary endpoint, was significantly lower after treatment with SER-155.

It should be noted that alloHCT recipients are severely immunosuppressed and at high risk of infectious complications. Transmission of multidrug-resistant organisms from stool bank products has led to fatal infections in immunocompromised patients, including one documented death at Massachusetts General Hospital of a patient who underwent HCT [85]. These led to a strengthening of FMT donor screening protocols by the health authorities' agencies and international recommendations have been issued for FMT donor screening [86]. It underscores the continued importance of careful evaluation of FMT safety, in particular with respect to infectious complications [87].

Future research should focus on optimizing the use of selected bacterial consortia versus whole microbiota ecosystems, donor selection, administration methods (e.g., oral capsules, duodenoscopy or colonoscopy/enema), and on the long-term safety to maximize probiotics therapeutic potential in alloHCT settings. In addition, for prospective trials testing interventions to maintain or increase microbiome diversity, the choice of the primary clinical endpoint to prove the intervention efficacy is a challenge. Given intestinal dysbiosis appears to play a role in the pathogenesis of acute GVHD, the primary endpoint may be acute GVHD, with non-relapse mortality as a secondary endpoint. Alternatively, using overall survival as the primary endpoint would likely provide the strongest evidence to determine whether maintaining or restoring microbiome diversity translates into a clinical benefit.

Postbiotics refer to a preparation of inanimate microorganisms and/or their components [88]. Clinical data are very scarce with only the results of a phase II study available as a congress abstract [89]. Twenty patients ingested sodium butyrate (600 mg $\times$ 2 capsules per day) from engraftment (day +14 to +21) until days +90 to +100 and were compared to an historical control group, with a significantly lower incidence of grade III-IV aGVHD in the postbiotics group. Randomized studies are needed to confirm these findings.

CONCLUSION AND FUTURE DIRECTIONS

The gut microbiome is critical to post-transplantation immune reconstitution, and its diversity and composition have been linked to alloHCT outcomes, particularly the development and severity of aGVHD and the risk of opportunistic infections. Disruption of microbial diversity—by conditioning regimens, antibiotics, or

environmental factors—has been consistently associated with increased GvHD-related morbidity and impaired immune recovery. Clinical trials exploring microbiome-targeted strategies—such as prebiotics, FMT and microbial-derived therapies, and postbiotics—are promising but require confirmation, standardization and long-term safety data. Ultimately, integrating microbiome modulation into personalized transplant strategies has the potential to reduce complications, enhance immune recovery, and improve survival after alloHCT.

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AUTHOR CONTRIBUTIONS

DW, MT, MH, HP and FM wrote, provided critical review, editorial comments, and final approval of the manuscript.

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ADDITIONAL INFORMATION

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